Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Negotiate, Then Solve: Multi-Step LLM Agents | *Python, Gemini, RAG, Scikit-learn, CP-SAT*

- Built an agentic workflow that turns free-text requests from email, chat and a web portal into typed constraints, then negotiates conflicts over multiple rounds, adapting to each stakeholder reply via typed tool calls.
- Designed dual-process parsing: a calibrated TF-IDF + logistic-regression classifier fast-paths confident requests, while an LLM drafts typed constraint JSON validated deterministically.
- Built hybrid BM25 + dense-embedding retrieval with reciprocal-rank fusion over OCR-ingested policy documents, raising Recall@5 to **93.4%** (vs 87.7% BM25), with every denial required to cite a retrieved rule.
- Built an evaluation harness with baselines, oracle and ablations (McNemar, Wilcoxon, bootstrap CIs); guardrails ensured **no attack succeeded** across 50 prompt-injection, spoofing and data-leak tests.

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, FlashAttention-2, RoPE*

- Pretrained 25M-param decoder-only Transformers from scratch on 1B+ tokens across Hindi and Assamese using bare PyTorch primitives, driving a +36.5% perplexity reduction via FlashAttention-2, RoPE, and SwiGLU.
- Trained dedicated 16k SentencePiece BPE tokenizers on AI4Bharat Sangraha, eliminating token fertility inflation and achieving 0.505 Bits-Per-Byte sequence compression across both high and low-resource languages.
- Fine-tuned 8 model variants across 20k synthetic multi-hop reasoning tasks, unlocking a +45.8% relative Token F1 gain and 61.3% decomposed CoT step score over direct SFT via intermediate deductive scratchpads.

Football Video Analytics with YOLO | *Python, YOLOv8, ByteTrack, OpenCV, Streamlit, K-Means*

- Fine-tuned YOLOv8 with ByteTrack for multi-object tracking of players, referees, and the ball, leveraging deterministic K-Means clustering on jersey pixels for team classification and linear interpolation for ball-gap recovery.
- Compensated camera pan, tilt and zoom with chained RANSAC pitch-plane homographies over Lucas-Kanade flow, mapping camera pixels to pitch coordinates to compute real-world sprint velocities, distance covered, and team ball possession.
- Re-architected pipeline to stream frames in batches, **cutting peak RAM from 5.3 GB to 1.0 GB** and full-video decodes from six to two; built Streamlit app with content-hashed caching for instant re-runs and stats export.

Distributed Data Pipelines: MapReduce & gRPC Streaming | *Hadoop, C++, Python, gRPC*

- Reduced a 1M-record Hadoop analytics job (19 outputs) to a single MapReduce stage by expressing every metric as a mergeable partial aggregate, so only compact summaries cross the shuffle; results match sequential and MPI baselines exactly.
- Architected a real-time gRPC streaming pipeline sustaining **96–123k records/s**, answering mid-ingestion queries in 2.1–4.5 ms with bounded-queue backpressure and zero record loss.

Technical Skills

Languages: Python, C/C++, SQL, Bash.

ML & Statistics: Logistic Regression, K-Means Clustering, TF-IDF, Hybrid Retrieval, Hypothesis Testing, Bootstrap CIs.

Deep Learning & LLMs: PyTorch, Transformers, Hugging Face, LoRA, RAG, LLM Agents, YOLOv8, OpenCV.

Data & Tools: NumPy, Pandas, Scikit-learn, Hadoop MapReduce, gRPC, FastAPI, Streamlit, Linux, Git, Docker.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.